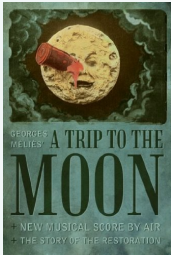


(1) Multi-Channel Retriever

Target sample: one available modality

A group of men travel to the moon by being shot in a capsule from a giant
.....

Or

Text-only Image-only

Text/Image Encoder

Target Emb x_q

Memory Bank

Superman and Supergirl take on the cybernetic Brainiac,



Class:"Animation" ,"Action", "Adventure"

Similarity Retriever Top-K

Retrieved Instances

Class

Image

Text

Retrieved Emb r_i

(2) Differentiated Feature Extraction

Target Emb

Retrieved Emb

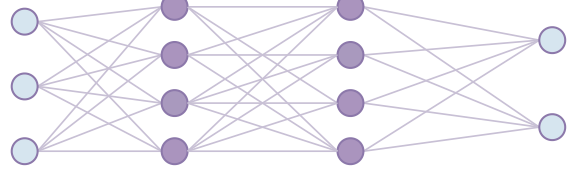
x_q

r_i

Target – Retrieved difference

Retrieved Diff Emb

SAE



Sparse difference bottleneck



Z_{Target}
Self soft prompt



$Z_{\text{Retrieved Diff}}$
Difference soft prompt

(3) Label Attention Mechanism

K: Retrieved Emb

Q: Target Emb

r_i

x_q

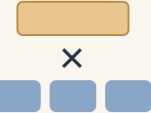
Cosine

Scale

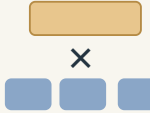
Softmax

Mean

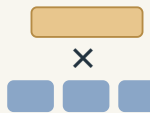
Attention Score



Label1



Label2



Label n

Retrieved Instance Class

Label Attention

Label-attention soft prompt

Shared inputs: x_q = target embedding r_i = retrieved embedding, $i = 1, 2, \dots, k$

Prompt assembly and prediction

Available observation

Candidate labels

Self soft prompt

Difference soft prompt

Label-attention soft prompt

Qwen2-VL
Multimodal large language model

Label + Explanation

Final prompt

Text/Image-only